A1 Q2a Edward Tanurkov 20401544

Wednesday, September 30, 2026 4:17 PM

Showing weak stationarity

a) $X_t = (-1)^t Z_0 + (-1)^{t+1} Z_1$

$= (-1)^t Z_0 - (-1)^t Z_1$

$= (-1)^t (Z_0 - Z_1)$

Need to show μ does not depend on t

$E[X_t] = (-1)^t (E[Z_0] - E[Z_1])$

$= (-1)^t (0 - 0)$ 0 mean by defn

$= (-1)^t (0)$

$= 0$

$\therefore$ constant μ $\forall t$ ✓

Need to show $Cov(X_t, X_{t+h}) = \gamma(h)$ does not depend on t

$\gamma(h) = Cov(X_t, X_{t+h})$ ↓ 0 from above

$= E[X_t X_{t+h}] - E[X_t]E[X_{t+h}]$

$= E[X_t X_{t+h}]$

$= E[[(-1)^t (Z_0 - Z_1)][(-1)^{t+h}(Z_0 - Z_1)]$

$= E[(-1)^{t+h}(-1)^t (Z_0 - Z_1)^2]$ $(-1)^{2t+h} = (-1)^h$ b/c $(-1)^{2t} = (-1)$ $\forall t$

$= E[(-1)^{2t+h}(Z_0^2 + Z_0 Z_1 + Z_1^2)]$ $(-1)^h$ constant in terms of t, pull out of Expec.

$= (-1)^h (E[Z_0^2] + E[Z_0]E[Z_1] + E[Z_1]^2)$ $E[Z_0 Z_1] = E[Z_0]E[Z_1]$ by indep.

$= (-1)^h (\sigma^2 + 0 + \sigma^2)$

$= 2\sigma^2 (-1)^h$

$\therefore$ no dependence on t $\forall t$ ✓

Thus, we've shown X_t is weakly stationary.

■

Calculating ACF

Above, $\gamma(h) = 2\sigma^2 (-1)^h$

$\gamma(0) = 2\sigma^2$

$$\rho(h) = \frac{\gamma(h)}{\gamma(0)} = \frac{2\sigma^2 (-1)^h}{2\sigma^2}$$

$\therefore \rho(h) = (-1)^h$ is the ACF for X_t.

A1 Q2b Edward Tanurkov 20401544

Monday, October 5, 2026 10:49 PM

Showing weak stationarity

b) Show $E[X_t]$ has mean not dep. on t

$$E[X_t] = E[Z_0\cos(ct)] + E[Z_1\sin(ct)]$$
$$= \cos(ct)E[Z_0] + \sin(ct)E[Z_1]$$
$$= 0 + 0$$
$$= 0$$

$\therefore$ constant $\forall t$ ✓

Showing $\gamma(h)$ does not dep. on t

$$\gamma(h) = \text{Cov}(X_t, X_{t+h})$$
$$= E[X_t X_{t+h}] \quad (\text{since } E[X_t] = 0, \text{ same reason as 2a})$$
$$= E[[Z_0\cos(ct) + Z_1\sin(ct)][Z_0\cos(c(t+h)) + Z_1\sin(c(t+h))]]$$
$$= E[Z_0^2\cos(ct)\cos(c(t+h)) + Z_0Z_1\cos(ct)\sin(c(t+h))$$
$$+ Z_0Z_1\sin(ct)\cos(c(t+h)) + Z_1^2\sin(ct)\sin(c(t+h))]$$

(simplifying the Z_0Z_1 terms $= 0$ b/c $E[Z_0Z_1] = 0$ by indep. and def'n, and $E[Z_0^2] = E[Z_1^2] = \sigma^2$ by def'n)

$$= \sigma^2\cos(ct)\cos(c(t+h)) + \sigma^2\sin(ct)\sin(c(t+h))$$
$$= \sigma^2[\cos(ct - (c(t+h)))]$$
$$= \sigma^2\cos(ct - ct + ch)$$
$$= \sigma^2\cos(ch).$$

$\therefore \gamma(h)$ does not dep. on t $\forall t$

Thus, we have shown X_t is weakly stationary.

■

Calculating ACF

$\gamma(h) = \sigma^2\cos(ch)$ as above

$$\gamma(0) = \sigma^2\cos(c0)$$
$$= \sigma^2$$

$$\Rightarrow \rho(h) = \frac{\sigma^2\cos(ch)}{\sigma^2}$$

$\therefore \rho(h) = \cos(ch)$ is the ACF of X_t

A1 Q2c

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Showing weak stationarity

c) Show mean does not dep. on t

$E[X_t] = E[Z_0 S_t] + E[Z_1 S_{t-1}]$

$= S_t E[Z_0] + S_{t-1} E[Z_1]$

$= 0 - 0$

$= 0$

$\therefore$ constant $\forall t$ ✓

Show $\gamma(h)$ does not dep. on t

$\gamma(h) = E[X_t X_{t+h}]$ (same reasoning as a), b))

$= E[[Z_0 S_t + Z_1 S_{t-1}][Z_0 S_{t+h} + Z_1 S_{t-1+h}]]$

$= E[Z_0^2 S_t S_{t+h} + Z_0 Z_1 S_t S_{t-1+h} + Z_0 Z_1 S_{t-1} S_{t+h} + Z_1^2 S_{t-1} S_{t-1+h}]$

$(E[Z_0 Z_1] = 0,\ E[Z_0^2] = E[Z_1^2] = \sigma^2$ by def'n)

$= \sigma^2 S_t S_{t+h} + \sigma^2 S_{t-1} S_{t-1+h}$

$= \sigma^2 [S_t S_{t+h} + S_{t-1} S_{t-1+h}]$

We have two cases to consider.

Consider h is even.

$t + h = t$ b/c period = 2

$\Rightarrow \sigma^2 [S_t S_t + S_{t-1} S_{t-1}]$

$\Rightarrow \sigma^2 [S_t^2 + S_{t-1}^2]$

But given period = 2, S_t and S_{t-1} will always have the same value. Let $a = S_t^2 + S_{t-1}^2$ for some $a \in \mathbb{R}$.

$\Rightarrow \sigma^2 a$ for some constant a.

Now, consider h is odd

t and t+h are different parity. Thus, S_t and S_{t+h} are different values

Thus, let $a = S_t S_{t+h}$ and $b = S_{t-1} S_{t-1+h}$ for constants $a, b \in \mathbb{R}$

$\Rightarrow \sigma^2 [a + b]$

We can do this b/c a and b each cover the entire period (S_t even, S_{t+h} odd).

$\therefore$ We have shown in all cases that $\gamma(h)$ does not depend on t given the parity produces some constants $a, b \in \mathbb{R}$ across the period of 2.

$\therefore$ We have shown X_t is weakly stationary.

■

Computing ACF

$\gamma(h) = \begin{cases} \sigma^2 a & \text{for even } h \\ \sigma^2 (a+b) & \text{for odd } h \end{cases}$ as per above

$\gamma(0) = \sigma^2 [S_t^2 + S_{t-1}^2] = \sigma^2 a$ as per above

$$\Rightarrow \rho(h) = \begin{cases} \frac{\sigma^2 a}{\sigma^2 a} & \text{for an even } h \\ \frac{\sigma^2(a+b)}{\sigma^2 a} & \text{for an odd } h \end{cases}$$

$$\therefore \rho(h) = \begin{cases} 1 & \text{for an even } h \\ \frac{a+b}{a} & \text{for an odd } h \end{cases} \quad \text{is the ACF of } X_t.$$

A1 Q2d Edward Tanurkov 20401544

Tuesday, October 6, 2026 11:51 AM

Showing weak stationarity

d) Show mean does not depend on t

$E[X_t] = E[Z_t Z_{t-1} \dots Z_{t-k}]$

$= E[Z_t]E[Z_{t-1}] \dots E[Z_{t-k}]$

$= 0 \cdot 0 \cdot \dots \cdot 0$

$= 0$

$\therefore$ constant $\forall t$

Show $\gamma(h)$ does not depend on t

$\gamma(h) = E[X_t X_{t+h}]$ (same reasoning as a, b, c)

$= E[(Z_t Z_{t-1} \dots Z_{t-k})(Z_{t+h} Z_{t-1+h} \dots Z_{t-k+h})]$

$= E[(\prod_{i=0}^{k} Z_{t-i})(\prod_{i=0}^{k} Z_{t-i+h})]$

There are two cases to take care of here.

Let $a = \prod_{i=0}^{k} Z_{t-i}$

Let $b = \prod_{i=0}^{k} Z_{t-i+h}$

Case 1: $h = 0$

If $h=0$, then a and b are in the same set perfectly.

This is shown as:

$a = \prod_{i=0}^{k} Z_{t-i}$

$b = \prod_{i=0}^{k} Z_{t-i+h}$

$= \prod_{i=0}^{k} Z_{t-i}$

$= a$ exactly.

Thus: $E[ab] = E[Z^2_t Z^2_{t-1} \dots Z^2_{t-k}]$

$= \sigma^2 \sigma^2 \dots \sigma^2$ $k+1$ factors since we start at $i=0$

$= \sigma^{2(k+1)}$

Case 2: $h \neq 0$

If $h \neq 0$, then b/c b will have at least one Z_{t-i+h} value offset from a.

Example: $h=1$

$a = \prod_{i=0}^{k} Z_{t-i}$

$b = \prod_{i=0}^{k} Z_{t-i+1}$

There is no Z_{t-i} factor in b, thus

$E[ab] = [Z_t Z^2_{t-1} \dots]$

(First factor of a; First factor of b)

$= 0 \cdot \sigma^2 \cdot \sigma^2 \dots$

$= 0$

This is true for any value of $h \neq 0$ b/c it is guaranteed to produce this offset.

$\therefore$ We have shown that

$$\gamma(h) = \begin{cases} \sigma^{2(k+1)} & \text{for } h = 0 \\ 0 & \text{fo } h \neq 0 \end{cases}$$

which does not depend on t.

$\therefore$ We have shown that X_t is weakly stationary. ■

Computing ACF

$$\gamma(h) = \begin{cases} \sigma^{2(k+1)} & \text{for } h = 0 \\ 0 & \text{for } h \neq 0 \end{cases}$$

$$\Rightarrow \rho(h) = \frac{\gamma(h)}{\gamma 0}$$

$$= \frac{0}{\sigma^{2(k+1)}}$$

$\therefore \rho(h) = 0$ is the ACF of X_t.